

Name: _____

MATH 344

Spring 2026

Final Exam

Assume A, B are “generic” $n \times n$ matrices, T is a triangular $n \times n$ matrix, and $\mathbf{u}, \mathbf{v}, \mathbf{x}, \mathbf{b} \in \mathbb{R}^n$. In terms of floating point operations, determine the computational complexity of each of the “operations” below. Give your answer in terms of “big O”, e.g. $O(n^p)$ for some p . For credit, you must answer *all* of the below correctly.

- (a) _____ : Dot Products, $\mathbf{u} \cdot \mathbf{v}$.
- (b) _____ : Matrix-vector multiplication, $A\mathbf{v}$
- (c) _____ : Matrix-matrix multiplication, AB
- (d) _____ : Computing the RREF of A
- (e) _____ : Computing $\det A$ with cofactor expansion.
- (f) _____ : Solving $T\mathbf{x} = \mathbf{b}$ with “back/foward substitution”